



Unit 3 · Outcome 1 · Maintaining Biodiversity

Reading the shape of life from the rocks

KK04 VCE Environmental Science · Units 3 & 4 · Exam study summary

Digital Vector EnviroSci · single-topic mastery summary with worked good-vs-bad answers, exam traps, command words and practice.



Reading the shape of life from the rocks

Unit 3 · AOS 1 · Maintaining Biodiversity · KK04 Reading the shape of life from the rocks Biodiversity hasn't crept upward at a steady pace. The fossil record shows life climbing, crashing, and exploding again — five catastrophic mass extinctions and several bursts of rapid diversification . This is how environmental scientists read the rate and extent of that change straight out of layered rock. diversity rising mass extinction recovery / radiation drag to orbit · 540 million years of marine family diversity VCAA dot point 3.1.2.1 Inferred from the foss

The fossil record is a torn flip-book

How we know The fossil record is a torn flip-book Every claim in this dot point — that diversity rose, crashed, or exploded — comes from one source of evidence: the fossil record . Fossils form when dead organisms are buried in sediment that hardens into rock. Layer piles on layer, so the column of rock at Wirra Gorge is a stack of time. Analogy · the torn flip-book Imagine a flip-book animation of life on Earth. The fossil record is that book — but with most pages torn out . Soft-bodied animals (jellyfish, worms) rarely survive as fossils. Erosion sh

"Rate" and "extent" are not the same axis

The two words examiners reward "Rate" and "extent" are not the same axis The dot point asks about variation in rate and extent . Students who blur these together lose easy marks. They answer two different questions about any change in biodiversity. Rate How fast biodiversity is changing — species lost or gained per unit of time . A mass extinction is a spike in the rate of extinction. Steady, low-level loss is the background rate . Extent How much — the magnitude of the change, usually as a percentage of taxa affected. End-Permian's extent (~96% of marin

The "Big Five" mass extinctions

When the rate spikes The "Big Five" mass extinctions Palaeontologists Raup & Sepkoski (1982) plotted marine families through time and found five moments where the extinction rate rose far above background. A mass extinction is defined as the loss of roughly $\geq 75\%$ of species in a geologically short interval (typically under ~2 million years). 5 major events in 540 My ~96% peak extent (end-Permian) <2 My "geologically short" 10 My+ typical recovery time Tap a card to flip it. Front: when and how severe. Back: the cause and why it matters. The thread that li

Rapid diversification fills the empty world

When the rate spikes the other way Rapid diversification fills the empty world Extinction is only half the dot point. The fossil record also shows the opposite spike: rapid species diversification (also called adaptive radiation), where the number of species climbs steeply in a short interval. Two situations trigger it. Empty niches after a crash A mass extinction vacates ecological niches . Surviving lineages, now free of competitors and predators, radiate to fill the space. The classic case: mammals exploded in diversity after the non-avian dinosaurs

Life's diversity: a jagged climb, not a smooth ramp

Putting the curve together Life's diversity: a jagged climb, not a smooth ramp Stand back from 540 million years and a pattern emerges. Overall biodiversity has broadly increased , but the line is punctuated — sharp crashes (the Big Five and many smaller events) and steep recoveries. The rate and extent of change are anything but constant. That variability is the dot point. ~541 Mya · Cambrian explosion Diversification spike Most major animal phyla appear rapidly — the foundational radiation of complex life. ~444 Mya · End-Ordovician Mass extinction · ~8

Extinction or diversification?

Active recall Extinction or diversification? Drag each statement (or tap it on a phone) into the bin it describes. Then check your answers. These are the two phenomena named in the dot point — drilling the distinction is exam gold. Mass extinction 0 Rapid diversification 0 s

Seven traps that cost real marks

⚠ EXAM TRAPS

Where marks leak away Seven traps that cost real marks ! "The dinosaurs died at the end of the Triassic." No — the end-Cretaceous (K–Pg), ~66 Mya , killed the non-avian dinosaurs. The end-Triassic cleared the way for them. Mixing these up is the most common deep-time error. ! Treating "rate" and "extent" as the same thing. They're different axes — how fast vs how much . A question naming both wants both addressed. ! Saying the fossil record is complete. It's a biased, incomplete sample — soft bodies rarely fossilise, erosion destroys layers. Percentages

Command-word decoder

💡 COMMAND WORDS

Answer the verb that's actually asked Command-word decoder The same content earns different marks depending on the command word. Tap each to see what the examiner wants — and a worked sentence using this dot point. Identify Describe Explain Compare Analyse Evaluate

P·E·L writing trainer

📝 P·E·L WRITING

Structure that scores P·E·L writing trainer Extended-response marks come from structure as much as facts. Build your answer as Point → Evidence → Link . Type a response to the prompt below; the meters light up as your writing hits each element, and detected key terms appear underneath. Prompt: Explain how a mass extinction can lead to a period of rapid species diversification, referring to evidence from the fossil record. (4 marks) Point 0% A direct claim that answers the command word Evidence 0% Named events, %, mechanisms, fossil clues Link 0% Ties evi



The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

Practice questions — with weak & strong responses

✗ WEAK ANSWER

—

✓ STRONG ANSWER & MARKING

strong responses Each question shows a weak and a strong response so you can see why marks are earned. Open the marking guide, mark your own attempt honestly, and the dock tracks your total.



The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

Digital Vector · VCE Environmental Science study summary · Maintaining Biodiversity · KK04 · digitalvector.com.au
Worked good/weak answers illustrate exam technique — always check against current VCAA marking advice.